## Comparative Summary of Plant Species Detection Research Papers (2025-2026)

| Ref | Paper Title | Author(s) / Year | Journal Name | Month & Year of Publication | Methodology & Technology | Advantages | Limitations |
|---|---|---|---|---|---|---|---|
| 1 | **Overview of PlantCLEF 2025: Multi-Species Plant Identification in Vegetation Quadrat Images** | Martellucci, G. et al. (2025) | *arXiv preprint (CLEF 2025 Working Notes)* | September 2025 | Benchmark/challenge paper; vision-transformer (DINOv2-based) baseline models; weakly-labelled multi-label classification on 1.4M training images and 2,105 expert-annotated quadrat test images (~400 species). | Large, expert-annotated benchmark enabling standardized comparison across methods.<br>Supports large-scale ecological monitoring and biodiversity assessment. | Training data is single-species while test data is multi-species, creating a drastic domain shift.<br>An overview/benchmark paper, not a deployable identification system. |
| 2 | **Multi-Label Plant Species Prediction with Metadata-Enhanced Multi-Head Vision Transformers** | Herasimchyk, H. et al. (2025) | *arXiv preprint (LifeCLEF, CLEF 2025 Working Notes)* | August 2025 | Pre-trained DINOv2 ViT-B/14 backbone with multi-head (species/genus/family) classification, multi-scale tiling, dynamic threshold optimization, and bagging/Hydra ensembling. | Ranked 3rd on the PlantCLEF 2025 private leaderboard.<br>Taxonomic multi-head design and multi-scale tiling improve robustness to scale variation. | Multi-scale tiling and ensembling are computationally expensive.<br>Performance still constrained by the single-to-multi-species domain shift. |
| 3 | **An Evaluation of a Pre-Trained Transformer-Based Self-Distillation Model (DINOv2) for Cross-Domain Plant Species Identification** | Ong, C. et al. (2025) | *Neural Computing and Applications (Springer)* | August 2025 | Pre-trained self-distillation transformer (DINOv2) fine-tuned on the PlantCLEF2020 dataset (~320,000 herbarium and field images, 997 species) for cross-domain adaptation. | 17.7% accuracy improvement over prior best ensemble/Siamese triplet-loss methods (HFTL-ENS, OSM-ENS).<br>Strong feature-extraction capability bridges herbarium-to-field domain gap. | Relies on the older, comparatively smaller PlantCLEF2020 dataset.<br>Herbarium-to-field cross-domain accuracy, while improved, remains imperfect. |
| 4 | **A Pipeline to Compile Expert-Verified Datasets of Digitised Herbarium** | Arno, J. et al. (2025) | *Plants, People, Planet (Wiley)* | December 2025 | GBIF-sourced, Darwin-Core-standardized sampling and preprocessing pipeline combined | Reusable pipeline applicable to any Darwin-Core-compliant | Species-level accuracy (65-72% top-1) is notably lower than |

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| | **Specimens for Automated Plant Identification to Accelerate Taxonomy** | | | | with expert verification; deep learning genus- and species-level classifiers tested on Cyperaceae and Rhamnaceae. | image dataset.<br>Combines automated dataset generation with expert verification; genus-level top-5 accuracy of 88-92%. | genus-level accuracy.<br>Validated on a limited number of plant families only. |
| **5** | **Enhanced Plant Species Identification through Metadata Fusion and Vision Transformer Integration** | Javed, H. et al. (2025) | *Computers, Materials & Continua (Tech Science Press)* | 2025 | Vision Transformer (ViT) models fused with contextual metadata (environmental conditions, geographic location, phenological traits) via a multi-modal data-fusion strategy; ImageNet-based transfer learning. | High test accuracy (97.27%) and Mean Reciprocal Rank (0.9842).<br>Metadata fusion notably improves discrimination between morphologically similar species. | Depends on reliable contextual metadata (location, environment), not always obtainable.<br>Requires high-performance GPU resources (RTX 3090, 18 GB), limiting accessibility. |
| **6** | **Multi-Label Plant Species Prediction with Metadata-Enhanced Multi-Head Vision Transformers (duplicate entry in source list)** | Herasimchyk, H. et al. (2025) | *arXiv preprint (LifeCLEF, CLEF 2025 Working Notes)* | August 2025 | Same DINOv2 ViT-B/14 multi-head, multi-scale tiling and ensemble approach described in Ref. 2 (this title appears twice in the source file). | Same as Ref. 2 above. | Same as Ref. 2 above; retained here only because the source list repeats this entry. |
| **7** | **Herbify: An Ensemble Deep Learning Framework Integrating Convolutional Neural Networks and Vision Transformers for Precise Herb Identification** | Sheth, F. et al. (2025) | *Plant Methods (BioMed Central)* | July 2025 | Custom 'Herbify' dataset built via a Preprocessing Algorithm for Herb Detection (PAHD); ensemble of transfer-learned CNNs (EfficientNetV2-Large) and ViTs (ViT-Large/16). | Very high F1-score: 99.13% (EfficientNetV2-Large alone), 99.56% (ensemble).<br>Delivered as a working application ('Herbify') for practical herb identification. | Small dataset (6,104 images, 91 species, ~67 images/species) may limit generalization.<br>Narrow domain scope (herb species only). |

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| **8** | **Multi-Scale Transformer–CNN Fusion with Metric Learning for Medicinal Plant Identification** | Vullam, N. et al. (2025) | *IEEE Conference Proceedings (ICRCICN 2025)* | 2025 | Swin Transformer (global context) fused with EfficientNetV2 (local texture) via CBAM attention; ArcFace metric-learning layer; PCA for dimensionality reduction. | Outperforms conventional CNNs and standard transfer-learning approaches.<br>Effectively discriminates visually similar medicinal plant species. | Evaluated only on a small public Kaggle dataset covering 20 species.<br>Limited real-world/field validation beyond the benchmark dataset. |
| **9** | **Venation-Aware Hybrid CNN-Transformer for Fine-Grained Leaf Species Identification** | Hasan, M. R. et al. (2025) | *Bangladesh Journal of Botany* | December 2025 | Hybrid CNN-ViT architecture with edge/vein extraction, cross-attention fusion of local CNN features and global ViT tokens, and a venation-consistency objective; Grad-CAM/attention-rollout interpretability. | Very high performance: macro-F1 and balanced accuracy of 0.9973.<br>Robust to background and colour-shift variation; interpretable predictions. | Restricted to only 5 leaf species (banana, guava, jackfruit, mango, neem).<br>Relatively small dataset (2,500 images total). |
| **10** | **Long-Tailed Image Classification on the Specimens of Herbarium Dataset** | Ahmed, R. et al. (2025) | *IEEE Conference Proceedings (IJCNN 2025)* | June-July 2025 | Three-stage approach: pre-trained ViT feature extraction, parameter specialization for class imbalance, and residual fusion for unified predictions; evaluated in few-, medium-, and many-shot settings. | Reports state-of-the-art results on both Herbarium 2021 and Herbarium 2022 datasets.<br>Directly targets the long-tailed/rare-species imbalance common in botanical datasets. | Few-shot (rare) species remain harder to classify than many-shot (common) species.<br>Trained/evaluated on herbarium sheet images only, not natural field photos. |

# Research Gaps, SDG Alignment, and References

Based on: Comparative Summary of Plant Species Detection Research Papers (2025-2026)

## Research Gap Identification

### *1. Domain shift between training and deployment conditions*

Multiple papers (Refs 1, 2, 3, 6) show a persistent gap between single-species training images and multi-species/field test conditions, and between herbarium specimens and live field photos (Refs 3, 10). No paper fully closes this gap.

### *2. Narrow taxonomic and geographic scope*

Most high-accuracy systems are validated on small slices of the plant kingdom — 20 species (Ref 8), 5 species (Ref 9), 91 species (Ref 7), or 1–2 families (Ref 4). None demonstrate that these accuracy levels hold across thousands of species or diverse biomes simultaneously.

### *3. Dependence on unreliable or costly inputs*

Ref 5's metadata-fusion approach needs environmental/location metadata that isn't always available, and needs high-end GPU hardware (RTX 3090, 18GB) — a barrier to field or low-resource deployment.

### *4. Long-tailed/rare species underperformance*

Ref 10 explicitly shows rare/few-shot species are still classified less accurately than common ones — a gap relevant to biodiversity monitoring, where rare species often matter most for conservation.

### *5. Computational cost vs. practicality*

Ensemble and multi-scale tiling methods (Refs 2, 6) improve accuracy but are computationally expensive, raising questions about real-time or edge-device deployment (e.g., mobile apps for field botanists).

### *6. Lack of deployable, end-to-end systems*

Several entries are benchmark/overview papers (Ref 1) or evaluations of methodology rather than working applications. Ref 7 (Herbify) is a notable exception, but is narrow in domain (herbs only).

**Synthesized gap statement:** There is no single system in this literature that combines (a) broad taxonomic coverage, (b) robustness across herbarium-to-field and single-to-multi-species domain shifts, (c) resilience to rare-species imbalance, and (d) low computational/metadata requirements suitable for field deployment.

## Relevant SDG Goals

- **SDG 15 – Life on Land**: Core alignment; automated species ID directly supports biodiversity monitoring, habitat assessment, and conservation of terrestrial ecosystems.
- **SDG 2 – Zero Hunger**: Medicinal/crop plant identification (Ref 8) and rapid taxonomy tools (Ref 4) support agricultural biodiversity and food-relevant plant research.
- **SDG 3 – Good Health and Well-Being**: Medicinal plant identification (Refs 7, 8) supports pharmacological research and traditional medicine verification.
- **SDG 13 – Climate Action**: Large-scale vegetation monitoring (Refs 1, 4) contributes to tracking ecosystem change linked to climate impacts.

- **SDG 9 – Industry, Innovation and Infrastructure**: Development of AI/ViT-based identification pipelines represents technological innovation applicable to environmental infrastructure.
- **SDG 17 – Partnerships for the Goals**: GBIF-based, Darwin-Core-standardized pipelines (Ref 4) and open benchmark challenges (Ref 1) reflect global data-sharing collaboration among researchers.

## References


[1] Martellucci, G. et al. (2025). Overview of PlantCLEF 2025: Multi-Species Plant Identification in Vegetation Quadrat Images. arXiv preprint (CLEF 2025 Working Notes), September 2025.

[2] Herasimchyk, H. et al. (2025). Multi-Label Plant Species Prediction with Metadata-Enhanced Multi-Head Vision Transformers. arXiv preprint (LifeCLEF, CLEF 2025 Working Notes), August 2025.

[3] Ong, C. et al. (2025). An Evaluation of a Pre-Trained Transformer-Based Self-Distillation Model (DINOv2) for Cross-Domain Plant Species Identification. Neural Computing and Applications (Springer), August 2025.

[4] Arno, J. et al. (2025). A Pipeline to Compile Expert-Verified Datasets of Digitised Herbarium Specimens for Automated Plant Identification to Accelerate Taxonomy. Plants, People, Planet (Wiley), December 2025.

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[6] Herasimchyk, H. et al. (2025). Multi-Label Plant Species Prediction with Metadata-Enhanced Multi-Head Vision Transformers [duplicate entry in source list]. arXiv preprint (LifeCLEF, CLEF 2025 Working Notes), August 2025.

[7] Sheth, F. et al. (2025). Herbify: An Ensemble Deep Learning Framework Integrating Convolutional Neural Networks and Vision Transformers for Precise Herb Identification. Plant Methods (BioMed Central), July 2025.

[8] Vullam, N. et al. (2025). Multi-Scale Transformer–CNN Fusion with Metric Learning for Medicinal Plant Identification. IEEE Conference Proceedings (ICRCICN 2025), 2025.

[9] Hasan, M. R. et al. (2025). Venation-Aware Hybrid CNN-Transformer for Fine-Grained Leaf Species Identification. Bangladesh Journal of Botany, December 2025.

[10] Ahmed, R. et al. (2025). Long-Tailed Image Classification on the Specimens of Herbarium Dataset. IEEE Conference Proceedings (IJCNN 2025), June-July 2025.


Note: Reference 6 duplicates Reference 2 in the source table — flagged here since including both may inflate a formal citation count artificially.